

Optipeople DemoCNC D-2800M — Maintenance Manual

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This manual is for maintenance technicians. Operator-level cleaning is in [04-operator-manual.md §10](#). For any task marked **Service**, contact authorized service or open a ticket with support@optipeople.dk before proceeding.

1. Safety before any maintenance

⚠ Before any task that requires opening the enclosure, removing a guard, or working in the cabinet:

1. Park the machine and shut down via the HMI.
2. Turn isolator Q1 to OFF.
3. Apply your lock-out / tag-out device.
4. Bleed the pneumatic system: close V1, then open the bleed valve V2 on the cabinet underside until the gauge reads 0 bar.
5. Wait 5 minutes after power-off for the IPC capacitors to discharge.
6. Test that the spindles cannot be commanded by trying CYCLE START. The HMI must show "Power Off".

2. Maintenance schedule overview

Interval	Hours (typical 8 h/day, 5 d/week)	Section
Daily	every 8 h	§3
Weekly	40 h	§4
Monthly	160 h	§5
Quarterly	500 h	§6
Semi-annual	1,000 h	§7
Annual	2,000 h	§8
Spindle TBO	4,000 h	§9

The HMI shows a **Maintenance Due** badge when intervals come up; it also logs every maintenance action against the technician's account.

3. Daily tasks (operator can perform)

Task	Procedure
Visual check	Walk around. No leaks (oil/air), no loose panels, no foreign objects in the work zones
Vacuum table	Brush off all chips and dust. Inspect the gasket strip for tears
Dust drawer	Empty if > 1/2 full
Auto-grease level	HMI Status → Lubrication . Top up if < 25 %
Air filter bowl	Check FRL1 — drain if water visible, refill auto-oil reservoir if installed
HMI events	Skim the alarm log. Any repeated warning ≥ 3× this shift must be reported

4. Weekly tasks

#	Task	Notes
4.1	Linear guide wipe	Soft cloth with light oil along X-axis rails. Do NOT use compressed air
4.2	ATC magazine	Wipe slots, inspect pull-stud seats for chips
4.3	Top presser pads	Replace pads if compressed by > 30 % or surface is glazed
4.4	Side reference bar	Confirm pop-up cylinder cycles smoothly; lube the cylinder rod with PTFE spray
4.5	Vacuum gaskets	Visual inspection. Replace strip section if torn
4.6	Filter intake grille	Vacuum the cabinet intake grille; check fan operation
4.7	Dust hose clamps	Hand-test each clamp; re-tighten if loose
4.8	Spindle PT100 history	Open Diagnostics → Spindle Temps . No spindle should average > 65 °C

5. Monthly tasks

#	Task	Spec
5.1	Manual grease — X linear guides	OP-LUB-LGB grease, 2 strokes per nipple, 8 nipples total
5.2	Manual grease — Y/Z ball screws	OP-LUB-BSG grease, 1 stroke per nipple, 4 nipples
5.3	Manual grease — X rack	Thin film of OP-LUB-RACK across full rack length
5.4	Belt tension check (S3)	Plucked frequency 110 ± 5 Hz
5.5	Auto-grease pump output	Disconnect output line into a beaker; pump should deliver ≥ 0.3 cc/cycle
5.6	Pneumatic cylinder action	Cycle each manually via diagnostics; any cylinder slower than 0.5 s indicates seal wear
5.7	Cabinet door seals	Check seals close cleanly, no dust ingress
5.8	Earth bond	Megger test PE to chassis: < 0.1 Ω at 25 A



- OP-LUB-LGB — NLGI 2 lithium-complex grease (linear guide bearings)
- OP-LUB-BSG — NLGI 2 EP grease for ball screws
- OP-LUB-RACK — Adhesive open-gear lubricant
- OP-LUB-CYL — Pneumatic cylinder oil, ISO VG 32

Substitutes must match grade and base. **Do not** mix lithium-complex with polyurea greases.

6. Quarterly tasks (500 h)

6.1 Linear guide block re-grease cycle

Pump the auto-grease reservoir manually until each block oozes a small bead of grease. Wipe excess. This guarantees fresh grease has reached every block (auto-grease occasionally bypasses a saturated block).

6.2 Vacuum gasket strip replacement

Even with daily care, the foam gasket compresses and loses seal. Quarterly replacement is cheaper than fighting slipping parts.

1. Remove the old strip with a plastic scraper.
2. Clean the channel with IPA and a lint-free cloth.
3. Cut new strip (PN: OP-VAC-GAS-6X2) to length per channel.
4. Press in, butting joints tight. Do not stretch.
5. Run a vacuum hold-down test on a known-good panel.

6.3 Pneumatic FRL service

- Replace 5 µm pre-filter element (PN: OP-FLT-5UM).
- Replace 0.3 µm coalescing filter element (PN: OP-FLT-03UM).
- Inspect regulator diaphragm. Replace if puffing audibly at idle.

6.4 Spindle bearing thermal log

Export `spindle_temp_history.csv` from **Diagnostics** → **Export Logs**. A bearing trending up by > 5 °C month-over-month under the same load profile is an early-warning sign. Schedule \$9 inspection sooner.

6.5 Backlash measurement

Axis	Method	Pass
X	Dial indicator, 100 mm reversal, 5 cycles avg.	≤ 0.03 mm
Y	Same	≤ 0.02 mm
Z	Same	≤ 0.02 mm

Exceeding pass values: schedule \$7.3 pinion adjust or ball-nut preload.

7. Semi-annual tasks (1,000 h)

7.1 Auto-grease reservoir refill

The pump reservoir (PN: OP-GRP-2L) holds 2 L. Top up with OP-LUB-LGB through the fill port. Bleed any air via the pump's bleed nipple.

7.2 Brake test (Z gravity brake)

1. Park, then put machine in SETUP mode.
2. **Diagnostics** → **Drives** → **Brake Test**.
3. The HMI commands the Z drive to release the brake and apply a 1.2× rated holding torque. The Z position must not drift more than 0.05 mm in 10 s.
4. Failure: replace brake disc (PN: OP-BRK-Z01) — **Service**.

7.3 X-axis pinion preload

The X rack-and-pinion uses a master-and-slave pinion configuration. The slave is preloaded against the master to eliminate backlash. Preload slackens over time.

1. SETUP mode. Lift X servo cover.
2. Loosen the slave-pinion clamp by 1/2 turn.
3. Use the supplied torque wrench (PN: OP-TQR-25) at the preload adjuster. Target 8 ± 0.5 Nm.
4. Re-tighten the clamp. Verify backlash per §6.5.

7.4 EtherCAT termination check

Inspect each EtherCAT connector for moisture or bent pins. Reseat. Verify diagnostic LEDs are solid green at every slave.

8. Annual tasks (2,000 h)

#	Task
8.1	Replace auto-grease pump filter (PN: OP-GRP-FLT)
8.2	Replace all FRL elements (even if not visually clogged)
8.3	Drain and refill VFD cabinet filter media
8.4	Service the dust hood seals (replace if compressed)
8.5	Verify all 4 E-stop circuits and 3 door switches — see §10
8.6	Calibrate the laser tool setter against a master gauge ring (PN: OP-CAL-GR1)
8.7	Geometry survey — Service: laser interferometer X-axis, ballbar X-Y
8.8	Spindle vibration baseline — Service: vibration probe on each bearing
8.9	Export & archive the maintenance log to your CMMS

9. Spindle TBO (4,000 spindle-hours each)

Spindles are factory-rebuildable. At each spindle's individual 4,000 h mark (tracked per-spindle in **Maintenance** → **Spindle Hours**), schedule:

1. Spindle exchange — Service. Optipeople supplies an exchange unit; the old unit is sent for bearing replacement and recalibration.

2. Average lead time on exchange: 5 working days. Plan ahead. The HMI shows a yellow badge at 3,800 h and an orange at 3,950 h.

9.1 Symptoms suggesting early TBO

- Bearing temp > 80 °C sustained
- Audible whine that wasn't there last month
- Run-out at the collet face > 0.012 mm with a known-good holder
- Increased current draw at idle (VFD diagnostic)

10. Safety circuit verification (annual)

Tools: continuity meter, stopwatch.

1. E-stop test, each station:

- Press the E-stop.
- Time from press to spindle-stopped should be < 4 s for spindle running at 18,000 rpm.
- HMI must show E-9001 Emergency stop active .
- Verify both safety contactors K-SAFE-1 and K-SAFE-2 drop out (auxiliary aux contact reads NC).

2. Door switch test, each door:

- With machine cycling at moderate speed, open the door slightly until interlock breaks.
- Stop must occur within 1.5 s; HMI shows E-2010 .

3. Light curtain test (if fitted):

- Use the supplied test rod (PN: OP-LC-TST-14) — must trigger when placed anywhere in the protected plane.

4. Record results in the safety log. Failures must be rectified before returning the machine to service.

11. Consumables list

Part	Description	Typical life
OP-LUB-LGB	Linear-guide grease, 400 g cartridge	6 mo.
OP-LUB-BSG	Ball-screw grease, 400 g cartridge	12 mo.
OP-LUB-RACK	Open-gear lubricant, 1 kg	6 mo.
OP-VAC-GAS-6X2	Vacuum gasket strip, 5 m	3 mo.
OP-FLT-5UM / OP-FLT-03UM	FRL filter elements	3 mo. / 6 mo.
OP-PRSR-PAD	Top presser pad, set of 6	12 mo.
OP-BELT-PJ940	S3 spindle belt	12–18 mo.
OP-DUST-HOSE-200	Ø 200 mm flex hose, 1 m	24 mo.
OP-CAB-FAN-FLT	Cabinet intake filter	6 mo.
OP-IPC-BAT	IPC RTC battery (CR2032)	5 yr.

12. Maintenance log

Every maintenance action must be logged in the HMI under **Maintenance** → **New Entry**, with technician initials, hours-at-action, and any parts consumed. The log feeds the warranty / service record.

A monthly export should be archived in your CMMS or saved to the shared network drive. The HMI auto-exports on the first Sunday of each month if configured under **Settings** → **Integration** → **CMMS**.